EXCHANGE-RATE VOLATILITY AND FOREIGN PORTFOLIO INVESTMENT (FPI) FLOWS IN INDIA: AN EMPIRICAL INVESTIGATION

Course name: Applied Macro and Financial Econometrics (AMFE ODD 2025-26)

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ABSTRACT

The study investigates a high-frequency ARDL–ECM method to look into the macro-financial determinants of Foreign Portfolio Investments (FPI) coming to India, while at the same time using a VAR–Granger causality system to analyze the interaction direction between FPI and the major macroeconomic variables. Although earlier studies pointed out the factors behind capital flows in emerging markets, still the important questions regarding the dominance of long-run exchange-rate fundamentals on short-run market volatility, and the Indian FPI's symmetrically or asymmetrically responding to the domestic versus the global shocks, have not been answered. This study deals with these questions by means of equilibrium modeling, dynamic causality testing, and impulse-response analysis.

The ARDL outcomes support a long-run relationship that is stable, where the persistent currency depreciation is the most crucial factor exerting negative influence on FPI. The short-run dynamics depict the situation where global risk conditions, equity-market movements, and interest-rate differentials are very pronounced. The VAR–Granger framework indicates that the causation between exchange-rate changes and FPI is primarily one-sided, while impulse-response functions portray the long-lasting negative impact of currency shocks. The findings, when taken together, point out the dual influence of structural fundamentals and high-frequency disturbances on the capital-flow scenario of India.

Keywords: Exchange Rate, Foreign Exchange Volatility, Foreign Portfolio Investment(FPI), India, ARDL-ECM, VAR, Granger Causality.

1. INTRODUCTION

Foreign Portfolio Investment (FPI) is an important factor in India's capital markets, using the term to describe global investor sentiment and to a large extent affecting domestic liquidity and asset pricing. Financial intermediaries define FPI as the money non-residents put into financial assets like shares and bonds, which have a high liquidity and are prone to fast reversals. The latest figures show that foreign portfolio investors withdrew ₹7,945 crore from the Indian stock market in September 2025 only, which totaled the net outflow for the year so far to more than ₹1.4 lakh crore, mainly caused by global uncertainties and geopolitical tensions. These substantial movements in portfolio investments draw attention to the fact that foreign investors are constantly reshaping their views on India's macro-financial situation—not only through long-term growth fundamentals but also via daily volatility.

The sharp fluctuations in returns reveal the necessity to comprehend the factors behind the global investors' entering or exiting Indian markets. Although earlier studies have looked into the effect of currency volatility or market conditions separately, it is still very rare to find India-centered research of the high-frequency type that combines together the elements of currency risk, differential interest rates, accumulation of reserves, and world risk sentiment in determining foreign portfolio investor (FPI) flows. There is still a lot of discussion about whether the FPIs merely react to macroeconomic factors or if they intentionally cause currency fluctuations as a result. This research work, which employs daily-frequency ARDL-ECM framework along with VAR-Granger, impulse-response modeling, has. It ponders: How do the long-term exchange-rate determinants compare with the short-term currency shocks in impacting FPI? Is there an exchanging of causality between FPI and exchange rate or is it rather one-sided? These are the questions that lie at the heart of capital-flow-risk management, thus, they lead us to the literature review and empirical strategy that follows.

2. LITERATURE REVIEW

Due to the sophisticated and numerous-sided appearance of portfolio flows towards developing countries, the existing writings have to be reviewed very carefully since they can provide the understanding of the channels through which macro-financial forces interact with foreign investors' behavior. This part of the paper, taking more than a dozen key studies as a starting point, delves into important topics like exchange-rate fluctuations, global risk appetite, monetary policy and basic market conditions. These studies together not only set the ground for theoretical and empirical consideration of India's FPI dynamics but also point out the long-lasting gaps—especially concerning the long-run impact of currency movements and the direction of causality between macro-financial variables and portfolio flows—mainly regarding the Indian case. By bringing the controversies around these issues together, the review marks the present study's realm over the larger exchange of ideas and points out the still unanswered issues that incite our empirical research.

Dua and Sen (2006) conducted a similar study looking into the same relationships for an earlier period (1993-2004) through the use of quarterly data. They did not only use ADF, PP, and KPSS tests but also Johansen cointegration, VECM modelling, and dynamic diagnostic tools and thus established long-run cointegration among the real exchange rate, capital flows, their volatility, money supply, government expenditure, and the current account surplus. They found that every determinant Granger-causes the REER and that long-run impulse responses are in accordance with economic theory. Engaging in variance decompositions, net capital inflows and their volatility were pinpointed as the major determinants, followed by the fiscal and external sector variables.

Dua and Sen (2013) analyzed the impact of capital flows—mainly FPI and FDI—on the real exchange rate and its volatility by utilizing Indian quarterly data from 1993 to 2010. They used a range of techniques like DF-GLS stationarity tests, Johansen cointegration, VECM estimation and dynamic aids such as generalized impulse responses and variance decompositions to conclude that all variables were co-integrated and each was a Granger-cause of the real exchange rate. They demonstrated that net capital inflows and their volatility were the most significant determinants of the REER, with government spending, money supply, and current account following closely behind. Thus, FDI shocks were found to have negligible impact while FPI shocks led to the appreciation of the currency and a reduction of the volatility, whereas negative shocks resulted in increased volatility for a few quarters.

Garg and Dua (2014) did an analysis on long-term macroeconomic factors influencing the portfolio inflow to India using ARDL cointegration techniques. They took monthly data from 1995 to 2011. The authors looked at FPI, FII and ADR/GDR flows separately and employed the GARCH models to calculate exchange-rate volatility. The authors' findings endorsed the presence of long-run relationships and disclosed that domestic stock market performance, exchange-rate appreciation, and output growth are the main attractions for FPI and FII/ADR/GDR flows. Exchange-rate volatility was a strong disincentive for FPI and FII inflows, but it did not have an impact on ADR/GDR flows. This was a clear indication of the differences between structural and speculative investments. Granger causality further corroborated the influence from macroeconomic variables to portfolio flows.

Caporale et al. (2017) scrutinized the impacts of equity and bond portfolio inflows on exchange-rate volatility in the case of seven Asian emerging economies by relying on the data spanning from 1993 to 2015 (monthly data). By employing GARCH and Markov-switching models, they demonstrated that the inflows of capital into stocks are the major cause of volatility, while on the contrary, bond purchases are the main factor for the decrease in the aforementioned volatility. The regime-switching outcomes indicated that equity inflows raise the chances of either moving into or staying in high-volatility regimes, whereas bond inflows are the main support for low-volatility regimes. The results drew attention to a remarkable country-specific impact and associated high-volatility regimes with significant crises, such as the Asian and global financial crises.

Makoni (2020) conducted a study that examined the impact of real exchange rates and capital openness on Foreign Portfolio Investment (FPI) in nine African economies after the global financial crisis (2009–2016). The author applied panel fixed-effects models and accounted for heteroskedasticity and multicollinearity in his analysis, concluding that the depreciation of the real exchange rate discourages inflow of FPI. Furthermore, he argued that capital openness by itself cannot attract portfolio investments unless supported by good macroeconomic fundamentals. Among the factors influencing FPI positively, institutional quality, GDP growth, and stock market development were cited, highlighting the importance of economic stability and good governance.

Gautam et al. (2020) published an article in the Transnational Corporations Review in which they closely examined the influence of various capital flows, namely FPI, FDI, and debt flows, on the real effective exchange rate of the BRICS countries during the period from the first quarter of 1994 to the second quarter of 2019. Through the application of various econometric techniques like the ADF and PP tests, ARDL cointegration methods, and Granger causality analysis, the researchers concluded that, among other things, the portfolio inflows had a strong impact in the direction of real appreciation, hence, reducing external competitiveness. It was found that trade openness and foreign exchange reserves were the long-run determinants of REER with relatively stronger impacts whereas the effects of FDI were rather weak. The error-correction model showed slow rates of adjustment to the long-term equilibrium prevailing across the BRICS nations, thus pointing to the presence of stable long-run dynamics that were moderated by policy variables.

Prabheesh et al. (2022) reconsidered in MethodsX the interrelation of FPI flows and stock market performance in India during the COVID-19 pandemic, employing daily data from December 2019 to April 2021. Utilizing the Toda–Yamamoto causality approach and splitting the data into high-uncertainty and post-peak periods, the researchers concluded that during the early, uncertain phase of the pandemic, there was bi-directional causality between FPI flows and stock returns. However, after the peak, the causality turned unidirectional—from FPI flows to stock returns—thus, indicating that uncertainty regimes determine the nature of financial market interactions.

Bamidele (2023) investigated through a study on SSRN the influence of exchange-rate volatility on capital inflows from foreign portfolio investments (FPI) in Nigeria over a long period of 1986 to 2023. The study utilized ARDL, VAR with impulse response analysis, descriptive statistics, and GARCH modeling, and employed secondary data sourced from national and international institutions. It was affirmed that there were significant long-run relationships among the key variables. The study highlighted that previous FPI inflows and exchange-rate volatility were the primary factors determining the current level of FPI. It was observed that while immediate volatility possessed an insignificant negative impact on FPI, its lagged counterpart demonstrated a significant positive effect, thus indicating that investors take time to adapt to the fluctuations in the exchange rate. The research also pointed out GDP growth as a significant positive factor and verified the model's reliability through rigorous diagnostic testing. Moreover, it was able to establish a causal link between past volatility and current FPI inflows.

Jacob et al. (2023) investigated the exchange rate and foreign portfolio investment (FPI) in India with their dynamic relationship, and the influence of the macroeconomic factors such as foreign direct investment (FDI), industrial production, imports, interest rates, and inflation. With the help of stationarity tests, Johansen cointegration analysis, and a Vector Error Correction Model, the study detected three long-run cointegrating relationships and concluded that increased FPI inflows normally fortify the rupee. On the other hand, imports cause the exchange rate to go down, while interest rate differentials draw in capital and make the currency stronger. The case for inflation is ambivalent—moderate inflation turned out to be in favor of stability, while high inflation made it difficult to maintain stability. Capital flows were found to have a strong impact on exchange-rate behaviour both in the short and long run as the authors indicated.

Faizi et al. (2024), in their article for European Economic Letters, examined how exchange-rate volatility affected foreign investment inflows into India after reforms in the 1990s up to 2020. By applying a series of statistical tests including the Augmented Dickey-Fuller test for stationarity, the Engle-Granger method for cointegration in the long run, a VAR framework for the short-run dynamics, and the Wald tests for significance of the coefficients, the authors were not able to provide any proof of long-run cointegration between foreign investment inflows and exchange-rate volatility. Nevertheless, the short-run results presented a significant relationship which means that currency fluctuations do affect the behavior of investors by the time being. The research paper suggests that stability in exchange rates is a key factor in attracting and retaining foreign investments and also points out that even though long-term FDI inflows are stable, speculative capital movements still increase the volatility which particularly occurs in India's open financial market.

Sharma et al. (2024) published their study in The Review of Finance and Banking, where they discussed the causal relationships between foreign portfolio investments, service trade, currency value, and economic growth in India for the period 2012-2022. They conducted a series of tests, including ADF, KPSS, and PP for stationarity, ARDL bounds testing for long-run cointegration, and the Toda-Yamamoto Granger causality method, followed by extensive diagnostic and stability checks. Based on this, the authors found a one-way causation flowing from FPI to service trade and GDP and vice versa from service trade to foreign currency reserves. They additionally concluded that the service trade was not a major contributor to FPI in the short or long term, with economic growth and foreign reserves being the main positive influencers. On the other hand, an increase in the exchange rate led to a decrease in FPI.

Patra et al. (2025) published in the RBI Bulletin analyzed the effectiveness of the interventions made by the Reserve Bank of India (RBI) in the foreign exchange (FX) market as a tool to reduce the volatility of the exchange rate that was caused by the inflow and outflow of foreign investors due to the global financial situation. The authors also tried to find out if the impact of spot and forward interventions was different and if the effects of purchases and sales were alike. By using intraday event-study techniques, volatility and variance decomposition methods, and threshold-based nonlinear analyses, the authors presented that the Indian rupee's volatility in the recent past has been primarily caused by the volatility of the portfolio flows, which was mainly due to the global spillovers. The results of their study indicated that both USD/INR spot and forward interventions have reduced the volatility of capital flows significantly, with purchases and sales affected symmetrically, and are more effective in managing the volatility of the short term than the exchange-rate level alterations. They also pointed out that in the case of excessive short-term volatility, interventions are more effective, thereby indicating the need for synchronization of FX intervention strategies with the use of macroeconomic policy tools.

An analysis of the already published studies indicates that despite the fact that there are a lot of researchers investigating the interaction between capital flows and the movements of exchange rates in emerging markets, they mainly take a cross-country or regional approach and use lower frequency data, thus leaving a little country specific and high frequency evidence for India. Earlier studies have mainly concentrated on either the impact of fluctuations in exchange rate volatility and capital market movements or the spillovers of risk from the global markets, but the number of studies that have looked at all these factors together in one empirical analysis is very small.

But additionally scholars hold different views on the matter of whether or not foreign portfolio investments help cause volatility in exchange rates this is especially true in the case of India where there is not much agreement among the experts. Our review also points out that the majority of the existing studies do not apply a single methodology that combines ARDL-ECM for long run relationships and Newey-West Regression Model to capture short run predictive dynamics using daily data. The gaps outlined above are what prompt the current research that is looking into the role of exchange rate and its volatility in determining the extent and the direction of FPI flows into India, whether FPI flows contribute to the movement or the volatility of the Indian rupee and to what extent the interaction of domestic and global financial market variables with exchange rate dynamics may affect portfolio flows. Thus this paper will try to answer the following questions (i) What is the effect of the exchange rate and its volatility on foreign portfolio investment flows both in short and long run? (ii) Is the causality between exchange rate fluctuations and FPI flows bidirectional or unidirectional? (iii) How do the domestic indicators like NIFTY returns and Indian volatility index along with global indicators like global volatility and interest-rate differentials shape the FPI. The study, therefore, addresses a conspicuous gap in the literature by providing an India-based, daily-frequency, multi-method analysis illustrating the joint impact of currency risk and financial-market uncertainty on foreign portfolio flows.

3. DATA DESCRIPTION

Our examination is centered on daily time-series data whose range stretches from April 1, 2015, to March 29, 2024, which amounts to a total of 2,348 observations for each variable. The dataset is composed of a broad array of both domestic and global financial indicators that are pertinent to India's portfolio flows, which are obtained from various institutions like NSDL, RBI, MOSPI, NSE, US Treasury, CBOE, and Investment.com. The used variables are specified below.

I. FPI (IND_FPI)

The foreign portfolio investment (FPI) net inflow to India which was formerly monthly available in INR crores per day was converted with the help of the inverse hyperbolic sine function to handle extreme values and to keep it easily interpretable. The monthly figures obtained from NSDL and RBI were transformed into daily frequency by evenly distributing the monthly inflows over all trading days during the month. This variable is the dependent variable in the study because it directly mirrors the flow of foreign investors reacting to local and international conditions.

II. FOREX

The RBI reported India's foreign exchange reserves which were in INR crores underwent natural logarithm transformation to variance stabilization and proportional changes capturing. Since the data is weekly, it was daily series using linear interpolation converted which is rail because forex reserves are the smoothest evolving over time. The variable is included in the study as macroeconomic stability indicator because higher reserves mitigate the country's vulnerability to currency crises and also helps in building foreign investor confidence.

III. GLOBAL_VIX

The Global VIX which stands for global equity market volatility is a metric measured in percentage points and comes from the CBOE VIX index. Linear interpolation was used to fill in the gaps thus ensuring continuity and maintaining the smoothness of the series. This variable captures the global risk aversion which is an important factor influencing portfolio flows and thus the global uncertainty usually leads to a less risk-taking by foreign investors which in turn affects FPI movements in the emerging markets like India.

IV. IND_VIX

The India Volatility Index, which is also known as India VIX, is obtained from NSE and is expressed in percentage points. It indicates the level of uncertainty in the domestic stock market. The missing values in the dataset were processed by linear interpolation. The variable is added to illustrate domestic risk sentiment; a higher volatility in Indian markets might turn away foreign investors, thus affecting the daily FPI inflow.

V. GDP_GRW

India's GDP growth rate per quarter was originally reported as a percentage of growth and was converted into percentage points for consistency. The data was obtained from MOSPI/RBI, and it was uniformly distributed across all trading days in each quarter to form a daily frequency series. GDP growth is considered as a control variable reflecting the macroeconomic environment since good growth prospects usually lure foreign portfolio investment.

VI. IRD (Interest Rate Differential)

The Interest Rate Differential is the yield on India's 10-year government security minus the US 10-year Treasury yield calculated in percentage points based on data from RBI and the US Treasury. The monthly IRD rates were allocated to all trading days of that month. IRD represents a crucial macro factor because large differentials make Indian assets more desirable thereby, the chances of foreign capital inflow increase.

VII. CPI (Inflation)

The Consumer Price Index (Year to Date Inflation), acquired from MOSPI, and assessed in percentage points, was turned into a daily series by giving every trading day in that month the monthly value. CPI is taken into account to minimize the effect of inflation in the home market, which might affect the net returns of foreign investors and thus, the FPI flows.

VIII. NIFTY_50

NIFTY 50 symbolizes daily stock exchange returns and was converted from raw index values to daily log returns based on the data provided by the NSE. Missing days were interpolated and the resultant series is represented in percentage points. This variable reflects the local stock market trend, as a good performance or sudden volatility in the stock market can heavily influence the foreign investor's behavior.

IX. FX_VOL

Exchange rate volatility, determined as the 30-day rolling standard deviation of daily INR/USD exchange rate returns, is represented as a decimal but after transformation is shown in percentage points. The daily exchange rate data was interpolated wherever necessary before computing the rolling volatility. This variable is key to the study as it simply shows the currency risk—higher FX volatility might discourage foreign portfolio investors on the ground that there will be increased uncertainty in returns.

X. LN_ER (Exchange Rate)

The daily exchange rate (INR per USD) from Investing.com is converted into percentage points for the sake of uniformity within the model. The specific transformation or economic role (“Reason for Inclusion”) is marked as TBD, but generally, exchange rate levels are introduced to reflect the immediate impact of currency changes on foreign investors’ decisions and the assessment of capital flows from abroad.

4. METHODOLOGY

4.1 Descriptive statistics

Refer to Table 1. from the “Figures and Tables” section.

The enumeration of statistics yielded enormous differences in the actions of India’s macro-financial variables, especially those most influencing Foreign Portfolio Investment (FPI). The extreme range and volatility in the amount of FPI show that the investor's sentiment and the global risk situations are the main causes of fluctuations in the portfolio flows. Conversely, basic indicators like the natural logarithm of the foreign exchange reserves (LN_FOREX), show inverse but strong stability, indicating that India has been accumulating foreign reserves steadily throughout the sample period. The variations in exchange rates measured through the natural logarithm of exchange rate (LN_ER) returns, are very small and limited to the area around zero, not signifying any permanent trend but

rather very mild shocks happening now and then. During the same period, however, FX_VOL underwent a more violent and irregular path, with surges marking the times when there was uncertainty in the currency market—this is the main way the global risk aversion together with domestic macro-instability has been affecting the FPI behavior.

Apart from these key variables, there are other additional factors playing an important role in the FPI movements. Domestic (IND_VIX) and global (GLOBAL_VIX) volatility indices are on the same page when it comes to giving signals of financial market stress; the former and the latter produce extreme peaks with values exceeding 80 during crisis periods. The market players suffering from such uncertainty might have to reverse their FPI flows abruptly since they would be moving the investments quickly towards safe assets. The GDP growth is also a very mixed case—this is partly due to high-frequency interpolation but also reflecting major economic disruptions such as the collapse of the economy due to COVID-19 pandemic—thus underlining the importance of macroeconomic conditions for capital flows. All together, the descriptive statistics characterize a highly volatile system where FPI is very much influenced by both domestic macro-financial conditions.

4.2 Correlation

Refer to Figure 1. from the "Figures and Tables" section.

The correlation matrix shows that the short-run foreign portfolio investment (FPI) movements (dLN_FPI) are very weakly and linearly associated with most macro-financial variables, which highlights the very volatile and sentiment-oriented character of foreign portfolio flows. The returns on exchange rate (dLN_ER) and price-level dynamics (CPI) do not correlate significantly with FPI, which implies that daily portfolio flows are more prone to reacting to sudden shocks or expectations instead of being gradually adjusted through macroeconomic factors. Likewise, the correlation between LN_FOREX (and hence FX stability) is very weak, which aligns with the fact that the accumulation of India's foreign reserves by the central bank follows long-term structural trends and does not depend on daily foreign investor activities. Therefore, the extremely low pairwise correlations reaffirm that FPI flows are noisy, high-frequency, and primarily influenced by market sentiment rather than instant shifts in macroeconomic fundamentals.

Hierarchical variables to represent unreliability and danger, specifically GLOBAL_VIX and IND_VIX, show a strong internal correlation of (0.73), which is a confirmation that during the hard times, both the global and local volatilities are moving together. Their direct correlation with FPI is, however, not that strong at the daily level, but still the economic significance is quite high: VIX indices reacting to the disparity in foreign investment flows and the relationship between them being better captured through dynamic modelling than through static correlation. The correlation between IRD and GDP_GRW is 0.45 which indicates that there are some macroeconomic linkages between interest rates and growth conditions, both of which indirectly influence investor's confidence. Thus, daily correlations appear small but the matrix still reveals that the variables which are transmitting risk, those which are moving with macro fundamentals and those which need further investigation through multivariate modelling to capture their effect on FPI in the short and long run are all identified.

4.3 Stationarity Check

Refer to Table 2. from the "Figures and Tables" section.

The outcome of the Augmented Dickey–Fuller (ADF) test was such that a mixed order of integration for the macro-financial variables was discovered, and this confirmed the necessity of a modelling framework that could handle both the I(0) and I(1) processes. The various high-frequency market variables including LN_FPI, LN_ER, FX_VOL, IND_VIX, GLOBAL_VIX, and NIFTY_50 are shown to be stationary in levels, which means that the short-run fluctuations in portfolio flows, exchange rate returns, and volatility measures return to a normal stable mean and do not go through a persistent stochastic trend. This is in line with the economic reasoning—foreign portfolio investments respond rapidly to market trends and consequently, there is currency volatility, but it does not last for a long period of time. The fact that LN_ER and FX_VOL are stationary also implies that the dynamics of the exchange rate, though changeable, will not be characterised by a very long term of explosive behaviour. The stationarity of LN_FPI being I(0) also supports the literature that shows that sometimes, especially in high frequency, fops tend to reverse rather than trend

In contrast, the macroeconomic fundamentals represented by the variables—CPI, IRD, and LN_FOREX—were identified to be non-stationary, exhibiting continuous long-run fluctuations. The case of LN_FOREX being I(1) is understandable, since foreign exchange reserves develop slowly with monetary policy and external balance changes instead of being capped around a certain average. In like manner, CPI and IRD are affected by slow-moving structural factors in the same way, i.e., through policy cycles, inflationary pressures, and interest-rate adjustments, thus making them non-stationary even at daily frequency. Most importantly, these I(1) variables are crucial determinants of foreign investor behaviour, since foreign investor confidence in the long run is determined by inflation stability, interest rate differentials, and reserve adequacy. The presence of the I(0) and I(1) series justifies the application of the ARDL bounds-testing approach which permits the estimation of long-run relationships without requiring that all variables have the same order of integration.

4.4 Multicollinearity

Refer to Table 4. from the "Figures and Tables" section.

The initial OLS regression, performed with stationary transformations in order to meet the assumptions of the classical linear model, gives a starting point to the explanation of the relationship between the macro-financial variables and the changes in the flow of FPI. The fluctuations in the exchange rate (LN_ER) are associated with FPI in a significantly negative way, which indicates that at the time of the outflow of foreign capital the Indian rupee might be under depreciation pressure. The FX_VOL variable also reflects a negative, highly significant impact, which suggests that the foreign investors consider currency fluctuations as a risk and to avoid that they withdraw their investments. Conversely, the shocks in the domestic price condition (D_CPI) have a small positive but significant effect, while the GDP_GRW variable is characterized by a rapid and strong negative association, which supports the view that the high-frequency changes in GDP often indicate the times of macroeconomic stress rather than the times of real improvement in fundamentals. Differenced LN_FOREX comes in positively with no significance, which is in line with reserves being adjusted slowly and not having a say in the daily flow decisions. The first coefficients, taken together, indicate that the short-term dynamics of FPI are mainly determined by market volatility, currency movements, and global risk factors but not by structural fundamentals.

In order to establish the trustworthiness of these predictions, a Variance Inflation Factor (VIF) analysis was carried out to measure the degree of multicollinearity between the independent variables. The VIF values for all factors were far below the traditionally accepted cutoffs, the majority of them being very close to 1, a clear signal of extremely low collinearity. The two variables showing a weak relationship are IND_VIX (2.54) and GLOBAL_VIX (2.20), this is due to the fact that the latter is a global version of the former and therefore there will be some overlap in their meanings. These two variables, however, are still below the critical levels so one doesn't need to worry. The variables in question for this analysis such as LN_ER, FX_VOL, NIFTY_50, and GDP_GRW possess very low VIF values of approximately 1, which means that their individual contributions to the overall estimate of the regression coefficient are exactly the same and there is no redundant information. Mean VIF of 1.39 corroborates the fact that the set of predictors used in the study is of good statistical quality. To sum up, the joint application of OLS and VIF diagnostics has given us a green light to proceed to ARDL modelling as the stability, independence, and interpretability of the included variables are the virtues we can have confidence in.

4.5 ARDL Framework

Refer to Table 5. from the "Figures and Tables" section.

The choice of the Autoregressive Distributed Lag (ARDL) framework as the main estimation method was based on the fact that the dataset variables showed a mixed order of integration—some were I(0) while others were I(1). The ARDL model fits well to such data conditions and allows for proper estimation of both short-run and long-run relationships. The Akaike Information Criterion (AIC) was used to determine the optimal lag lengths, thus providing a model that is dynamically well-specified. The ARDL Bounds Test (Pesaran, Shin, and Smith) [*Refer to Table 6. from the "Figures and Tables" section.]* was then applied to assess the existence of a long-run equilibrium relationship between FPI and its macro-financial determinants. The outcome of the test supported cointegration, which allowed the estimation of the corresponding Error Correction Model (ECM) to continue. The resulting ECM contained a negative and statistically significant error-correction term, which implied stable convergence to the long-run equilibrium, and this model the paper uses to trace the long-run dynamics of the variables. Detailed tables of the full ARDL and ECM estimations are presented in Table 7. of the "Figures and Tables" section.

In order to verify the economic reliability of the ARDL-ECM results, detailed post-estimation diagnostic checks were performed. Among these checks the Breusch–Godfrey LM test for serial correlation, Breusch–Pagan/Cook–Weisberg and White tests for heteroskedasticity, the Cameron–Trivedi IM test for joint heteroskedasticity and normality, and the Ramsey RESET test for model specification were included. The first diagnostics pointed out the existing heteroskedasticity and it was possible only with the Newey–West heteroskedasticity-consistent (HAC) standard errors re-estimation. This HAC-adjusted ARDL-ECM is the empirical model for short run dynamics, which is used for inference, and all intermediate diagnostic statistics and robustness checks are reported in Tables 8-10. in the "Figures and Tables" section.

4.6 VAR Granger

Even though the ARDL–ECM framework employs both the long-run equilibrium relationship and short-run adjustments in FPI dynamics, still it is silent on the direction of influence among the variables. In order to complement the ARDL results and to gain insight into the predictive interdependencies of the system, the research resorted to Vector Autoregressive (VAR) modeling and Granger causality tests. Since VAR requires stationary data, all variables were transformed into their stationary representations prior to estimation. Lag-length selection using FPE, AIC, HQIC, and SBIC unanimously indicated an optimal lag order of four, thus guaranteeing a sufficiently rich dynamic structure to capture interactions between foreign portfolio investment (FPI) and exchange-rate movements. The VAR system was found to be stable, with all eigenvalues lying within the unit circle, which in turn supported the validity of the subsequent causality analysis.

Standard diagnostic checks for VAR were carried out, which involved checking stability, evaluating autocorrelation, and checking for normal distribution of the residuals. The overall VAR structure, however, is still interpretable and the causality results are still robust despite the fact that some deviations from normality and mild autocorrelation observed residuals, which are typical for financial time series. The complete VAR outputs and comprehensive Granger causality tables are provided in Table 11. of the "Figures and Tables" section.

4.7 Impulse Response Function

Refer to Figure 2. from the "Figures and Tables" section.

In order to enrich the VAR–Granger causality analysis with the dynamic insights obtained, the research used Impulse Response Functions (IRFs) to depict the time-path of FPI's reaction to structural shocks in important macro-financial variables. The Granger tests reveal the presence of directional causality among the variables, but they do not provide a measurement of how significant or how long lasting these impacts are. Thus, IRFs function as an important addition since they let us explore the shock’s propagation mechanism—especially that of changing exchange rates—through foreign portfolio investments. The use of orthogonalized IRFs allows for shocks to be separated and made uncorrelated, which in turn facilitates an unambiguous understanding of the effect of an unexpected change in one variable on the others in the VAR system.

5. EMPIRICAL RESULTS AND FINDINGS

5.1 ARDL-ECM

Refer to Table 7. from the “Figures and Tables” section.

The ARDL–ECM estimation technique unambiguously points out the existence of a long-run relationship between Indian macro-financial environment and FPI flows.

$$\begin{aligned} LNFPI_t = {} & \alpha_0 + \phi_1 LN\ \ FPI_{t-1} + \beta_1 LN\ \ FOREX_t + \beta_2 FX\_\mathrm{VOL} + \beta_3 LN\ \ ER_t \\ & + \beta_4 LN\ \ ER_{t-1} + \beta_5 LN\ \ ER_{t-2} + \beta_6 CPI_t + \beta_7 IRD_t + \beta_8 GLOBAL\_\mathrm{VIX}_t \\ & + \beta_9 IND\_\mathrm{VIX}_t + \beta_{10} GDP\_\mathrm{GR}W_t + \beta_{11} GDP\_\mathrm{GR}W_{t-1} + \beta_{12} NIFTY50_t + \varepsilon_t \end{aligned}$$

The error-correction term (ECT = –0.0418, $p < 0.001$) is both negative and significant at a very high level, which means that 4.18% of the previous period's imbalance in FPI flows gets adjusted each day.

$$\begin{aligned} \Delta LN\_\mathrm{FPI}_t = {} & \lambda\, EC_{t-1} + \beta_1\, \Delta LN\_\mathrm{ER}_t + \beta_2\, \Delta LN\_\mathrm{ER}_{t-1} + \beta_3\, \Delta CPI_t + \beta_4\, \Delta IRD_t \\ & + \beta_5\, \Delta GDP\_\mathrm{GR}W_t + \beta_6\, \Delta GDP\_\mathrm{GR}W_{t-1} + \beta_7\, \Delta NIFTY\_50_t + \varepsilon_t \end{aligned}$$

Such a slow yet systematic adjustment has been made in the case of global portfolio investors who act gradually in response to the macro-financial shifts that are fundamental. The long-run coefficients are indicative of the clear prevailing influence of exchange-rate fundamentals on the flow of foreign portfolio investments over time. The exchange-rate level

(LN_ER) has a very large, negative and statistically significant coefficient (–1410.617, p = 0.008), which indicates that the constant depreciation of the Indian rupee is cutting down the inflow of FPI drastically because it causes higher currency-risk exposure besides giving lesser foreign-currency returns. On the other hand, exchange-rate volatility (FX_VOL) is located on the negative side but does not reach the significance level (p = 0.210), which implies that the investors' expectations are more in favor of the long-run currency trends than short-lived fluctuations. Foreign-exchange reserves (LN_FOREX) factor in the same way in defining the long-run as well as the short-run regimes (–3.609, p = 0.375), which means that reserve accumulation has not been a major driver in making portfolio investments less or more attractive in the long run.

The fundamentals of the equity market come out as the second main long-run driver. The coefficient for the NIFTY_50 index is huge, positive, and also very significant (381.32, p = 0.005), which means that the foreign portfolio capital takes place continuously due to the strong performance of Indian equities. On the other hand, risk-sentiment variables act differently: the effect of GLOBAL_VIX is slightly positive (p = 0.097), which means that the periods of global risk-on are favourable for the inflows, while the effect of IND_VIX is slightly negative (p = 0.083) which states that the high domestic volatility makes the investors stay away, though neither of the effects is strong enough to be the dominant one among the channels, i.e., the exchange-rate and the equity market. The macroeconomic fundamentals-CPI (p = 0.657), IRD (p = 0.722), and GDP growth (p = 0.109)-still have no impact in the long run and thus further support the notion that the foreign portfolio investors do consider the market-based indicators more than the slowly changing macroeconomic aggregates when deciding to invest in India.

As for short-run conduct, due to diagnostic testing, the ARDL-ECM coefficients cannot be interpreted directly since it showed the existence of heteroskedasticity and autocorrelation in the residuals. In the methodology, a Newey-West HAC regression was employed to take care of this issue. Consequently, the short-run patterns under discussion in this study are merely interpreted through HAC-corrected estimates which afford robust inference on high-frequency relationships.

5.2 Newey west regression

Refer to Table 9. from the "Figures and Tables" section.

In order to make certain of reliable inference contrary to heteroskedasticity and mild serial correlation that were discovered in the first ECM specification, the short-run ARDL–ECM model had to be re-estimated using Newey–West HAC standard errors. A lag length of 2 was chosen, which was in line with the ARDL(1,0,0,2,1,1,0,0,2,1) setup—where the ECM includes first-order differenced dynamics and allows for up to two periods of adjustment—and was also attributed to empirical stability found in the Newey–West variance estimates.

$$\begin{aligned}\Delta LN_{FPI,t} = \lambda\, EC_{t-1} &+ \beta_1\, \Delta LN_{ER,t} + \beta_2\, \Delta LN_{ER,t-1} + \beta_3\, \Delta CPI_t \\ &+ \beta_4\, \Delta IRD_t + \beta_5\, \Delta GDP\_\mathrm{GRW}_t \\ &+ \beta_6\, \Delta GDP\_\mathrm{GRW}_{t-1} + \beta_7\, \Delta NIFTY50_t + \varepsilon_t\end{aligned}$$

The use of higher lags (7 or 8) resulted in very similar coefficients but increased standard errors, thus confirming that the autocorrelation further than two periods does not add any significant new information. Therefore, the Newey–West (lag 2) correction can be considered the most theoretically coherent and efficient estimator for the daily-frequency system.

The adjustments in the short-run estimates helped to support the long-run equilibrium relationship and at the same time, they explained the nature of the changes at high frequencies. The term of error correction has kept negative and very significant (ECT = –0.0405, $p < 0.001$) which means that every day about 4% of the disequilibrium in FPI flows is corrected—which is in line with the gradual market adjustment way of working. The movements in the exchange-rate have shown a specific two-day overshooting pattern: a simultaneous depreciation has a significantly strong impact and leads to a considerable reduction of FPI flows (ΔLN_ER = –24.08, p = 0.019), whereas the previous day's movement brings back part of the effect (ΔLN_ER(–1) = 29.36, p = 0.006), which indicates short-run corrective trading and investor repositioning. Inflation shocks bring about an immediate increase in FPI (ΔCPI = 2.18, p = 0.039) looking at the very short run while on the other hand the interest-rate differential has a marginally significant negative impact (ΔIRD = –7.17, p = 0.055) which is in accordance with short-run carry-trade adjustments. GDP growth does not have a significant short-run effect, while daily stock returns (ΔNIFTY = 5.24, p =

0.050) attract FPI flows which is indicative of tactical investment behaviour. In a nutshell, the Newey–West–corrected results provide a stronger empirical basis for the ECM specification and the conclusion that short-run FPI dynamics are mainly influenced by exchange-rate shocks, inflation surprises, and stock-market movements, while long-run adjustments stay in place and are clearly defined.

5.3 VAR Granger

Refer to Table 11. from the "Figures and Tables" section.

The lag-selection exercise offers the basis for the narrative. The fact that all information criteria—FPE, AIC, HQIC and SBIC—converge on a four-lag specification, demonstrates very clearly that the interactions between foreign portfolio investment (FPI) and India's macro-financial variables take place over a few days rather than instantly. The VAR(4) system, with all eigenvalues safely inside the unit circle, has good behavior; thus, it assures stable impulse responses and at the same time permits the interpretation of causality to be meaningful.

After the model is estimated, the well-known story of emerging markets starts to reveal itself. The LN_FPI equation shows an amazing R^2 of 0.934, indicating that the foreign portfolio flows in Indian economy has been very well documented through the years and the global investors adjust their positions gradually rather than through abrupt reversals. On the other hand, exchange-rate movements (d_LN_ER) have a R^2 of 0.428 which is more modest—a common pattern for FX markets in big emerging economies where the daily exchange-rate changes are considerably influenced by a broad global information set that goes far beyond the domestic indicators and includes them as well.

The outcomes of the Granger causality test have revealed the most remarkable and at the same time the most closely related to the stylised facts in emerging-market finance. It has been shown that the changes in the exchange rate are the leading cause of the Granger inflows of FPI, and the joint test for d_LN_ER → LN_FPI has given $\chi^2 = 10.77$ (df = 4) and $p = 0.0293$. This finding represents a common pattern in the world of emerging markets: the foreign investors are always on the lookout for currency movements, and they often consider them as indications of the risk, monetary policy, and the likelihood of getting good returns.

Moreover, the second lag ($\chi^2 = 8.05$, $p = 0.0045$) is the one that powers this causality, which implies that there is a very short period of time before the information about the currency is fully passed on to the portfolio decisions—a behaviour that is indeed in line with the "wait-and-observe" approach that foreign investors very often use in the markets that are very volatile.

In the opposite direction, though, the narrative is not as strong. Even if LN_FPI exhibits a one-period impact on the exchange rate ($p = 0.0472$), this influence does not carry on through several lags; the overall test is not significant ($\chi^2 = 3.96$, df = 4, $p = 0.411$). This indicates that the sudden movements of the capital, either inflows or outflows, might have the effect of the immediate intraday currency adjustments, but FPI does not provide the exchange rate with the characteristics of its future course systematically. This type of relationship is the feature of robust emerging-market FX systems such as India's, where the combination of market microstructure, RBI interventions, and deepening market liquidity serves to counteract the feedback loops from portfolio flows to currency levels.

The evidence is further substantiated by a more extensive VAR system that includes ten macro-financial variables. Among all the predictors, only GDP growth ($\chi^2 = 11.05$, $p = 0.004$) and the exchange-rate level, LN_ER ($\chi^2 = 7.98$, $p = 0.0185$) were found to Granger-cause IND_FPI. This outcome once more coincides with the patterns reflected in the literature: the FPI of emerging markets is very much to the upside/downside of the growth prospects and the exchange-rate expectations, but it is surprisingly unaffected by the other issues around that period that were considered like volatility indices, NIFTY returns, CPI, or FX volatility—none of which, in fact, appeared significant in your system either. However, the joint test ($\chi^2 = 33.27$, df = 18, $p = 0.0155$) points to the conclusion that the macro-financial environment is, in fact, significant when considered collectively, even if the individual channels do not work in isolation.

The results, taken together, tell a consistent story that aligns with India's situation as a major emerging market: the movements of the exchange rate and the domestic growth fundamentals are the main indicators looking ahead to the foreign portfolio investors, while the FPI flows do not have a broad impact on the future currency behaviour. This asymmetric causality structure characterizes countries where currency fluctuations are a result of a mixture of global risk appetite and local macroeconomic expectations plus capital inflows responding to these signals but providing limited systematic feedback on the financial environment in general.

To summarize, the VAR evidence places India firmly within the emerging-market pattern where foreign portfolio investment reacts quickly—and often asymmetrically—to the movements in the currency and the growth signals, thereby, underlining the dominance of macroeconomic fundamentals in determining capital flow behaviour.

5.4 Impulse Response Function

Refer to Figure 2. from the "Figures and Tables" section.

The VAR(4) model, which was chosen by all the information criteria and confirmed stable by the eigenvalue diagnostics, serves as the foundation for the impulse response analysis which gives a dynamic, time-path interpretation of foreign portfolio investment (FPI) response to unanticipated exchange-rate fluctuations. The orthogonalized IRF's resolving of a one-standard-deviation structural shock to the first difference of the log exchange rate (d_LN_ER) allows the interpreter to view the reactions as the causal impact of a sudden currency depreciation, but only if contemporaneous interactions are held fixed.

The narrative depicted by the IRF is characteristic of the capital-flow behavior in developing markets like India.

In the immediate aftermath of the shock (steps 0–2), the IRF indicates a sharp and significant drop in FPI inflows. The response at step 2 is the most negative one and the 95% confidence band maintains for sure the exclusion of zero. This short lag in the materialization is in line with VAR results which pointed out that the second lag of d_LN_ER was the main factor for Granger causality ($\chi^2 = 8.05$, $p = 0.0045$). To put it simply, the depreciation-risk channel—sudden rupee weakness increases the foreign-currency losses of Indian assets, raises hedging costs, and signals heightened uncertainty—this is what the foreign investors' flight reflects.

In the medium horizon (steps 3–10), There is a continuous negative reaction. The impact is fading, but the FPI stays below the pre-shock level. The confidence interval becomes larger, but still suggests a significant downward push on the capital flows. Such persistence is typical in emerging markets where, after depreciation, foreign investors give up less and less their caution because of the "fear-of-further-depreciation" their behavior, the slowing down of the recovery of their valuations, and the time taken for macroeconomic or policy stabilisation. In the case of India, which has a capital market that is integrated globally but remains sensitive to currency fluctuations, the medium-term drag that persists is in line with the past FPI drawdowns during FX tensions.

By the long horizon (steps 10–20),The outcome of the response is a gradual approach towards zero, but this happens without a negative turn or an overshoot. The estimate of the point is still slightly negative, and the expanding confidence intervals underscore the increase in uncertainty regarding effects in the distant periods. The lack of any recovery means that the depreciation shocks do not create "bargain-hunting" inflows or contrarian reactions of capital markets. On the contrary, the impact of the initial currency shock vanishes slowly, and no reversal is indicated—this is a finding that supports the asymmetric causality discerned in the VAR, where movements in FX predict FPI flows but not vice versa.

When considered together, the IRF results really help the VAR and Granger-causality findings to tell a good story. The VAR pointed out that there was a strong one-way causation from the exchange rate changes to FPI (joint $\chi^2 = 10.77$, $p = 0.0293$), and the IRF illustrates this relationship in a dynamic way: the depreciation shocks set off immediate outflows, continue to bring about medium-term caution, and finally lead to slow long-run normalisation. The wider emerging-market literature supports this asymmetric pattern, in which currency risk rather than domestic asset performance alone is a primary determinant of portfolio flows. In India's case, these findings go a long way in persuading the investor that the exchange rate stability is a crucial factor in winning their confidence.

To sum up, the IRF shows that the shocks in exchange rates have a large and long-lasting negative impact on FPI that it supports the VAR-ECM proof and also the actions of foreign investors in emerging markets. Abrupt currency devaluation periods, for instance, first, scare away short-term capital, second, make medium-term investors unhappy, and third, do not even signal the existence of long-run compensatory reversals—this is the reason why currency management still is a major macro-financial concern for portfolio investment in India to be continued.

6. POLICY RECOMMENDATIONS

This research highlighted that the movements of the exchange rate—both long-term trends and short-term shocks—are indicative of main factors determining the Foreign Portfolio Investment in India. As FPI has shown a very strong and consistent adverse reaction to the exchange rate dropping, the government should consider stability in the exchange rate as the most important factor for attracting and retaining foreign investments. It would be necessary to increase transparency regarding foreign-exchange interventions, make the monetary policy easily understandable, and keep sufficient reserves to avoid the situation where investors with a long-term horizon will be driven away by prolonged depreciation.

Another important way that the strengthening of India's currency-hedging and derivative markets can be done. Although the volatility indices (GLOBAL_VIX and IND_VIX) show a gradual decline of effects in the long run, it is the short-run impulse response analysis that points out the sharp reactions of the investors to the sudden changes in the exchange rates. A more profound and easily accessible ecosystem of FX derivatives, particularly for foreign institutions, has the potential to diminish the currency-risk premiums, lessen the outflows during the stress periods that are panic-driven, and cause the investors' behavior to be more aligned with the fundamentals and less with the short-term volatility in the market.

Due to the considerable reaction of FPI to the future exchange rate changes expectations, the forward guidance and the communication strategy of the Reserve Bank of India (RBI) can be improved. Transparent declaration of monetary policy goals, economic outlook, and intervention methods can support the establishment of expectations and avert the disrupting self-fulfilling situations where the expected depreciation quickens the outflow of capital.

The research emphasizes the significance of macroprudential and capital-flow management tools as well. India has an open capital account for portfolio flows. However, during periods of extreme volatility, temporary measures may be necessary and targeted to the specific situation or where large and fast inflows or outflows pose a threat to the stability of the currency. The use of reserve buffers, liquidity provisions, and prudential norms in a flexible but rules-based manner can assist in easing the most destabilizing cycles without sacrificing the long-term objective of market openness.

Finally, considering the strong long-term positive correlation between the equity market performance and FPI, it is vital that the domestic market fundamentals are still strengthened. The policies that promote good corporate governance, increase the number of players in the market, make the market more liquid and draw in more institutional investors make India more attractive for global capital managers. Moreover, the constant supervision of world risk cycles particularly—the interplay among GLOBAL_VIX, IND_VIX and FPI flows—would enable the Indian government to predict the spillovers and synchronize the policy responses during the times of increased uncertainty more effectively.

In sum, the findings imply that the strengthening of the currency, promoting market depth, and controlling the global spillovers through coordinated, transparent policy actions would be the main conditions for the sustained and resilient portfolio inflows into one of the world's most dynamic emerging markets.

7. CONCLUSION

This research conducts a detailed analysis of the macro-financial factors that determine the process of Foreign Portfolio Investment (FPI) into India, and also points out the importance of particular aspects of literature that have been neglected so far regarding long-run exchange-rate fundamentals, high-frequency market volatility, and the dynamics of capital flows through their directionality. The usage of the high-frequency ARDL-ECM framework accompanied by the VAR-based Granger causality and impulse-response analysis has led to the discovery of a long-run equilibrium that is consistent and stable. In this equilibrium, the exchange rate - especially its long-lasting decline - proves to be the most potent structural barrier to portfolio inflows. On the other hand, the fluctuation of the short-run around the volatility indices, interest rates' differences, and global risk sentiment are only temporary effects, which is in line with the behaviour of foreign investors who are operating in an emerging market which is becoming more integrated yet still prone to uncertainties.

By revealing a primarily unidirectional causal pattern, the VAR analysis supports this interpretation: movements in the exchange rate and changes in GDP are the main factors seen through which FPI flows are predicted. Moreover, FPI does not take part in the changes of future exchange rates. The impulse-response functions indicate that, although, the currency shocks cause the FPI to decline right away and persistently, the latter is still considered as a trap of the exchange-rate instability in India during times of global stress.

When considered as a whole, the outcomes of the various studies point out the importance of macro-financial stability—foremost among them, the management of exchange rates—for the attraction of foreign portfolio investments. The combination of the aforementioned policies and the communication strategies of the Reserve Bank of India can greatly impact the risk-return profile of India for international investors, as the policies will limit continuous depreciations, broaden hedging and derivative markets, improve market liquidity and corporate governance. The communication strategies of the Reserve Bank of India, on the other hand, can greatly impact the risk-return profile of India for international investors. In a situation where capital flows are very responsive to both the structural fundamentals and the short-term disruptions, it is very important to keep the confidence in the management of the economy macro by the government in order to get the stable and resilient inflows of portfolios.

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Figures and Tables

Table 1. Descriptive Statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
LN_FOREX	2348	15.02488	0.2901281	14.56484	15.49899
LN_ER	2348	0.000124	0.0032342	-0.0196484	0.0231563
LN_FPI	2348	1.011566	7.102035	-9.282264	8.993569
FX_VOL	2348	4.747926	1.854716	0.9791051	12.72569
NIFTY_50	2348	0.0010413	0.0106387	-0.1390375	0.0840029
CPI	2348	5.259352	1.567067	1.083032	9.634551
IRD	2348	7.032849	0.5727825	5.81	8.02
GDP_GRW	2348	5.524361	6.274316	-23.9	20.1
IND_VIX	2348	17.1847	6.609112	10.135	83.6075
GLOBAL_VIX	2348	17.94039	8.079151	0	82.69

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

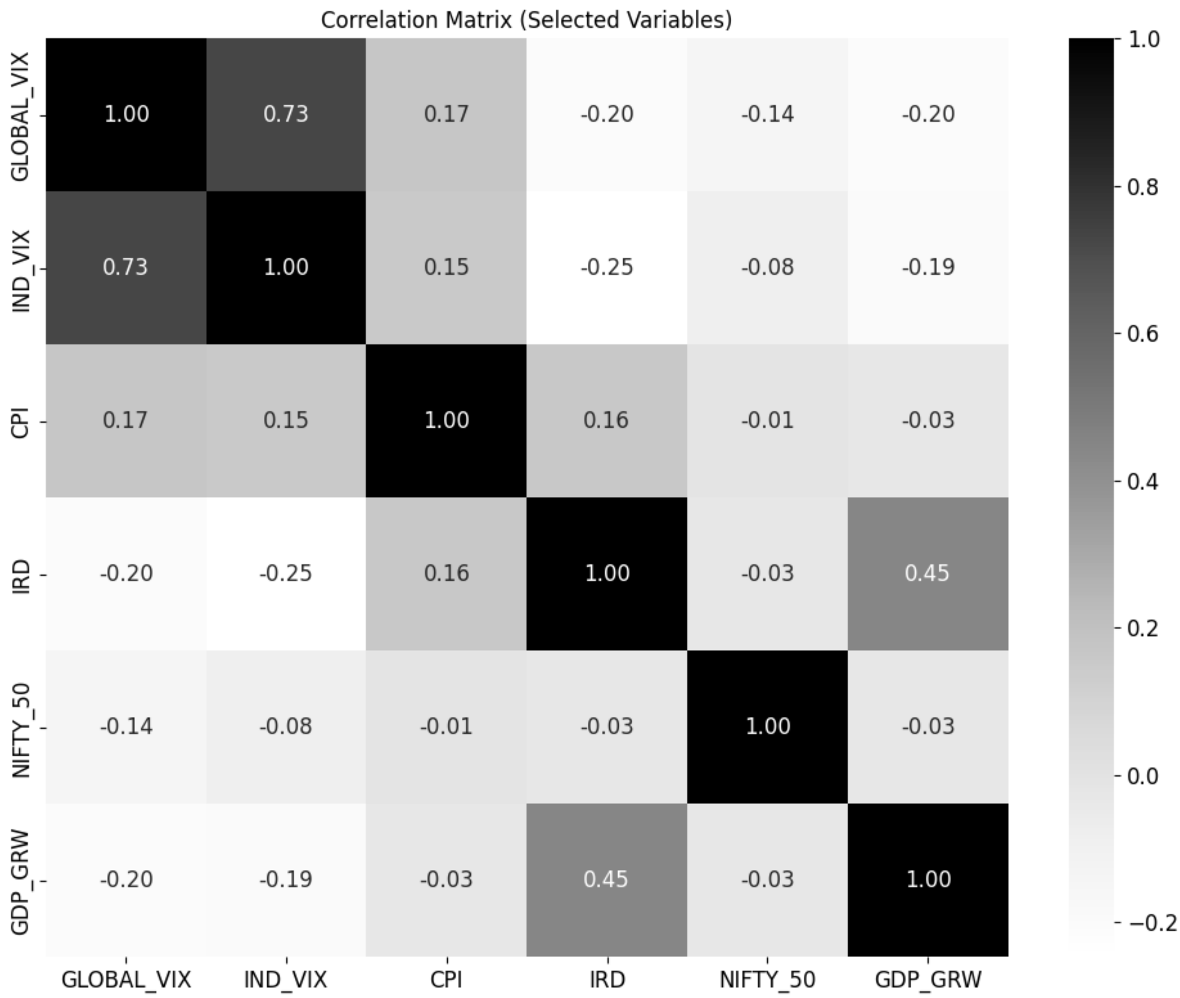

Figure 1. Correlation Matrix

Source: Author's own representation in Python by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 2. ADF Test Results

Null hypothesis: Not Stationary

Variable	ADF Statistic	5% Critical	p-value	H_0	Stationarity
LN_ER	–14.341	–2.860	0	Reject	I(0)
LN_FPI	–6.938	–2.860	0	Reject	I(0)
FX_VOL	–6.395	–2.860	0	Reject	I(0)
NIFTY_50	–14.078	–2.860	0	Reject	I(0)

CPI	–2.548	–2.860	0.1043	Fail to Reject	I(1)
IRD	–1.906	–2.860	0.3291	Fail to Reject	I(1)
GDP_GRW	–3.361	–2.860	0.0124	Reject	I(0)
IND_VIX	–5.519	–2.860	0	Reject	I(0)
GLOBAL_VIX	–4.974	–2.860	0	Reject	I(0)
LN_FOREX	-0.255	-2.860	0.9316	Fail to Reject	I(1)

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 3. OLS Results

Variable	Coefficient	Std. Error	t-Statistic	p-Value	95% Confidence Interval
LN_ER	-113.24	44.4193	-2.55	0.011	-200.3454 to -26.1347
D_LN_FOREX	51.4951	39.3744	1.31	0.191	-25.7172 to 128.7074
D_CPI	2.1452	0.8502	2.52	0.012	0.4780 to 3.8125
D_IRD	-11.9953	3.7468	-3.2	0.001	-19.3428 to -4.6479
GLOBAL_VIX	-0.1431	0.0251	-5.7	0	-0.1924 to -0.0939
IND_VIX	-0.131	0.033	-3.97	0	-0.1956 to -0.0663
GDP_GRW	-0.2601	0.0224	-11.61	0	-0.3041 to -0.2162
NIFTY_50	26.8857	13.6065	1.98	0.048	0.2036 to 53.5676
FX_VOL	-0.2904	0.0847	-3.43	0.001	-0.4566 to -0.1243
Constant	8.6069	0.4795	17.95	0	7.6666 to 9.5472

Statistic	Value
Observations	2,347
R-squared	0.1341
Adj. R-squared	0.1307

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 4. VIF Test Results

Variable	**VIF**	**1/VIF**
IND_VIX	2.54	0.3935
GLOBAL_VIX	2.2	0.4542
FX_VOL	1.32	0.7566
NIFTY_50	1.12	0.892
LN_ER	1.1	0.9055
GDP_GRW	1.06	0.9451
D_IRD	1.06	0.9478
D_CPI	1.05	0.9479
D_LN_FOREX	1.01	0.9931
Mean VIF	1.39	—

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 5. ARDL Results

ARDL(1,1,1,0,0,0,0,1,0,0) Estimates

Variable	Coefficient	Std. Error	t-Statistic	p-Value
LN_FPI (–1)	0.9597	0.0057	169.72	0.000 ***
CPI	2.1911	0.2288	9.58	0.000 ***
CPI (–1)	–2.1802	0.2287	–9.53	0.000 ***
IRD	–7.2529	1.0090	–7.19	0.000 ***
IRD (–1)	7.2108	1.0092	7.14	0.000 ***
LN_FOREX	–0.1467	0.1702	–0.86	0.389
IND_VIX	–0.0160	0.0093	–1.71	0.087 *
GLOBAL_VIX	0.0123	0.0075	1.64	0.102
GDP_GRW	–0.0091	0.0068	–1.35	0.178
NIFTY_50	5.5533	3.6749	1.51	0.131
NIFTY_50 (–1)	11.7670	3.5049	3.36	0.001 ***
FX_VOL	–0.0312	0.0253	–1.23	0.217
LN_ER	–23.2813	11.9324	–1.95	0.051 *
Constant	2.7209	2.8356	0.96	0.337

$R^2 = 0.9376$ & Adj. $R^2 = 0.9373$

F-statistic = 2695.78 ($p = 0.0000$) & N = 2,346

Note:
*** $p < 0.01$
** $p < 0.05$
$p < 0.10$

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 6. ARDL Bounds Test Results

Panel A: F-statistic Bounds

Test Statistic	Value	10% CV I(0)	10% CV I(1)	5% CV I(0)	5% CV I(1)	1% CV I(0)	1% CV I(1)
F-statistic	**7.631**	1.88	2.99	2.14	3.30	2.65	3.97

Panel B: t-statistic Bounds

Test Statistic	Value	10% CV I(0)	10% CV I(1)	5% CV I(0)	5% CV I(1)	1% CV I(0)	1% CV I(1)
t-statistic (ECM lag)	**–7.387**	–2.57	–4.56	–2.86	–4.88	–3.42	–5.54

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 7. ARDL-ECM Results

Dependent Variable: ΔLN_FPI

Section	Variable	Coefficient	Std. Error	t-Statistic	p-Value
Adjustment Term	ECT (LN_FPI(-1))	**–0.0418**	0.0057	–7.39	0.000
Long-Run Coefficients (LR)	LN_FOREX	–3.6093	4.0645	–0.89	0.375
	FX_VOL	–0.7607	0.6062	–1.25	0.210
	LN_ER	**–1410.617***	535.4046	–2.63	**0.008**
	CPI	0.2760	0.6206	0.44	0.657
	IRD	–0.7280	2.0469	–0.36	0.722
	GLOBAL_VIX	0.3119	0.1878	1.66	0.097

	IND_VIX	–0.3863	0.2226	–1.74	0.083
	GDP_GRW	–0.2580	0.1608	–1.60	0.109
	NIFTY_50	**381.3249***	137.2105	2.78	**0.005**
Short-Run Coefficients (SR)	ΔLN_ER	33.0598	17.0819	1.94	0.053
	ΔLN_ER(–1)	**29.8509**	11.5134	2.59	**0.010**
	ΔCPI	**2.1709***	0.2283	9.51	**0.000**
	ΔIRD	**–7.1135***	1.0084	–7.05	**0.000**
	ΔGDP_GRW	0.0039	0.0432	0.09	0.927
	ΔGDP_GRW(–1)	**0.1354***	0.0431	3.14	**0.002**
	ΔNIFTY_50	**–10.3934***	3.6785	–2.83	**0.005**
	Constant	2.7083	2.8286	0.96	0.338

Model Statistics:

Observations = 2346

R-squared = 0.0811

Adjusted R-squared = 0.0744

Log-likelihood = –4665.40

Root MSE = 1.7746

ARDL Model Selected by AIC: **ARDL(1,0,0,2,1,1,0,0,2,1)**

Source: Author’s own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 8. Diagnostic Tests Results

Diagnostic Test	Statistic	df	p-Value	Conclusion
Breusch–Godfrey LM Test for Serial Correlation (lag 1)	$\chi^2 = 0.517$	1	0.4722	No autocorrelation
Breusch–Pagan/Cook–Weisberg Test for Heteroskedasticity	$\chi^2 = 4.55$	1	0.0330	Evidence of heteroskedasticity
White's General Test for Heteroskedasticity	$\chi^2 = 1329.17$	167	0.0000	Strong heteroskedasticity present
IM-Test: Skewness	$\chi^2 = 81.75$	17	0.0000	Non-normal residual skewness
IM-Test: Kurtosis	$\chi^2 = 35.91$	1	0.0000	Excess kurtosis present
IM-Test: Overall Normality / Heteroskedasticity	$\chi^2 = 1446.83$	185	0.0000	Not joint normality and homoskedasticity

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 9. Newey-west Regression Results

Variable	Coef.	Std. Error	t-stat	p-value
ECM	-0.0405	0.00636	-6.37	0.000
d_LN_ER	-24.0809	10.2249	-2.36	0.019
ld_LN_ER	29.3645	10.5954	2.77	0.006
d_CPI	2.1815	1.0559	2.07	0.039
d_IRD	-7.1706	3.7332	-1.92	0.055
d_GDP	-0.00515	0.00575	-0.90	0.370

ld_d_GDP	0.1338	0.1828	0.73	0.464
d_NIFTY	5.2382	2.6735	1.96	0.050
Constant	2.6234	0.4176	6.28	0.000

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 10. Ramsey Reset Test Results

Statistic	Value	df	p-value
t-statistic	**1.4258**	2323	**0.1540**
F-statistic	**2.0330**	(1, 2323)	**0.1540**
Test SSR	6.3888	1	—
Restricted SSR	7306.478	2324	—
Unrestricted SSR	7300.090	2323	—

Null hypothesis (H_0): Model is correctly specified, no omitted variables, no functional-form misspecification
Decision rule: Fail to reject H_0 if $p > 0.05$

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

Table 11. Granger Causality Test Results

Dependent Variable: IND_FPI

Excluded Variable	Chi-sq	df	p-value	Granger Causality?
FOREX	0.0347	2	0.9828	No
GLOBAL_VIX	0.2076	2	0.9014	No
IND_VIX	0.4473	2	0.7996	No
GDP_GRW	**11.0526**	2	**0.0040**	**Yes**
IRD	0.1401	2	0.9324	No
CPI	0.0866	2	0.9576	No
NIFTY_50	4.2738	2	0.1180	No
FX_VOL	0.3713	2	0.8306	No
LN_ER	**7.9809**	2	**0.0185**	**Yes**
All variables jointly	**33.2700**	18	**0.0155**	**Jointly significant**

Source: Author's own calculations in Stata by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

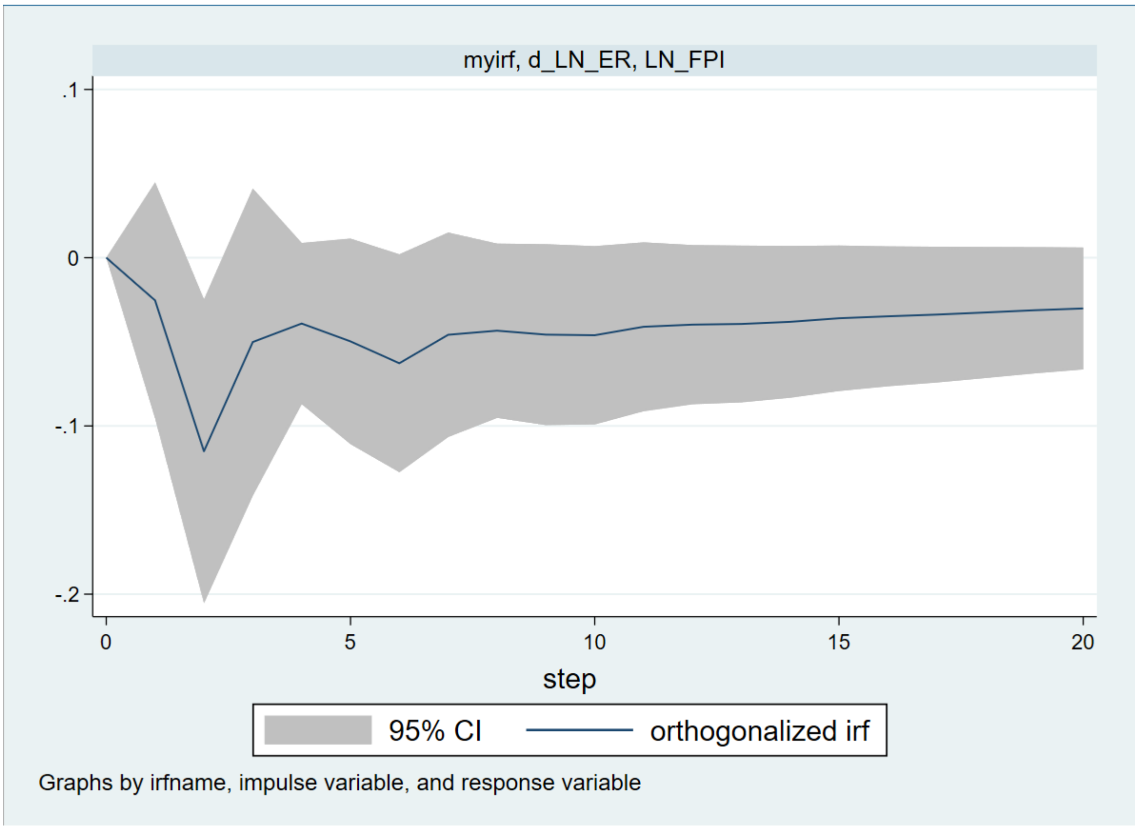

Figure 2. Impulse Response Function

Source: Author's own representation in Python by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

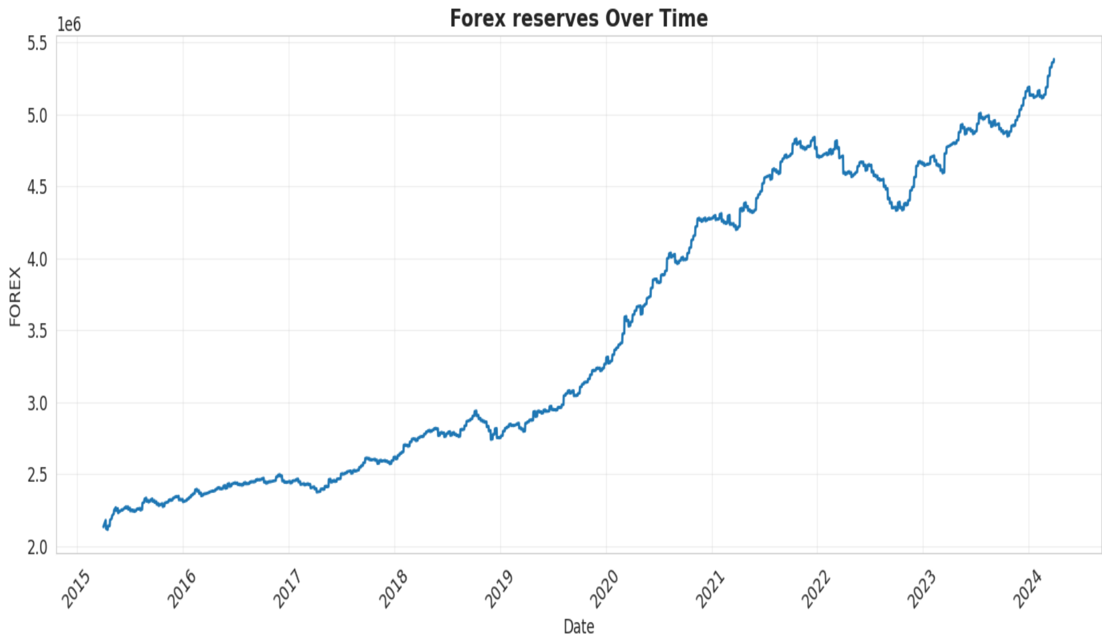

Figure 3. Foreign Exchange over time

Source: Author's own representation in Python by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

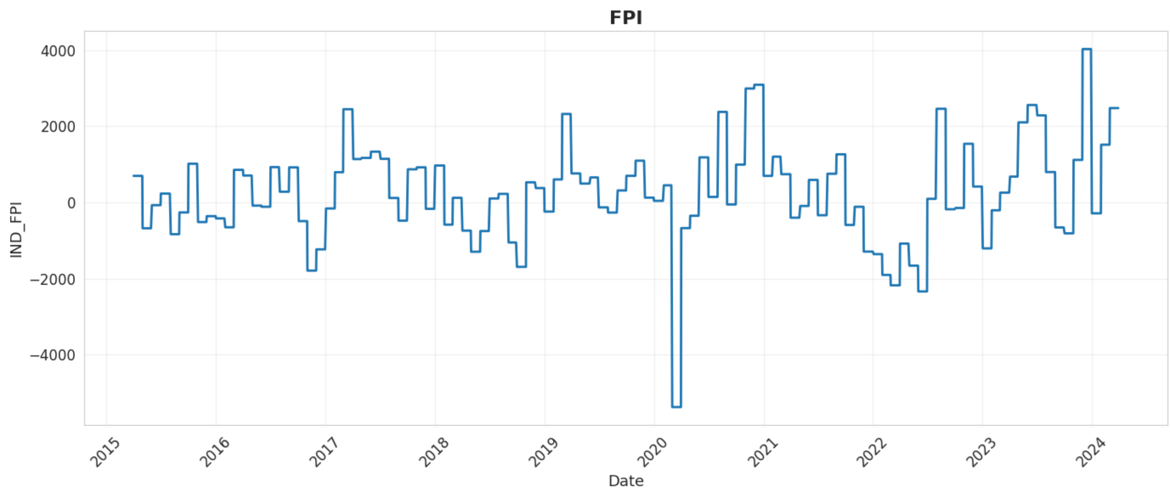

Figure 4. Foreign Portfolio Investment over time

Source: Author's own representation in Python by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024

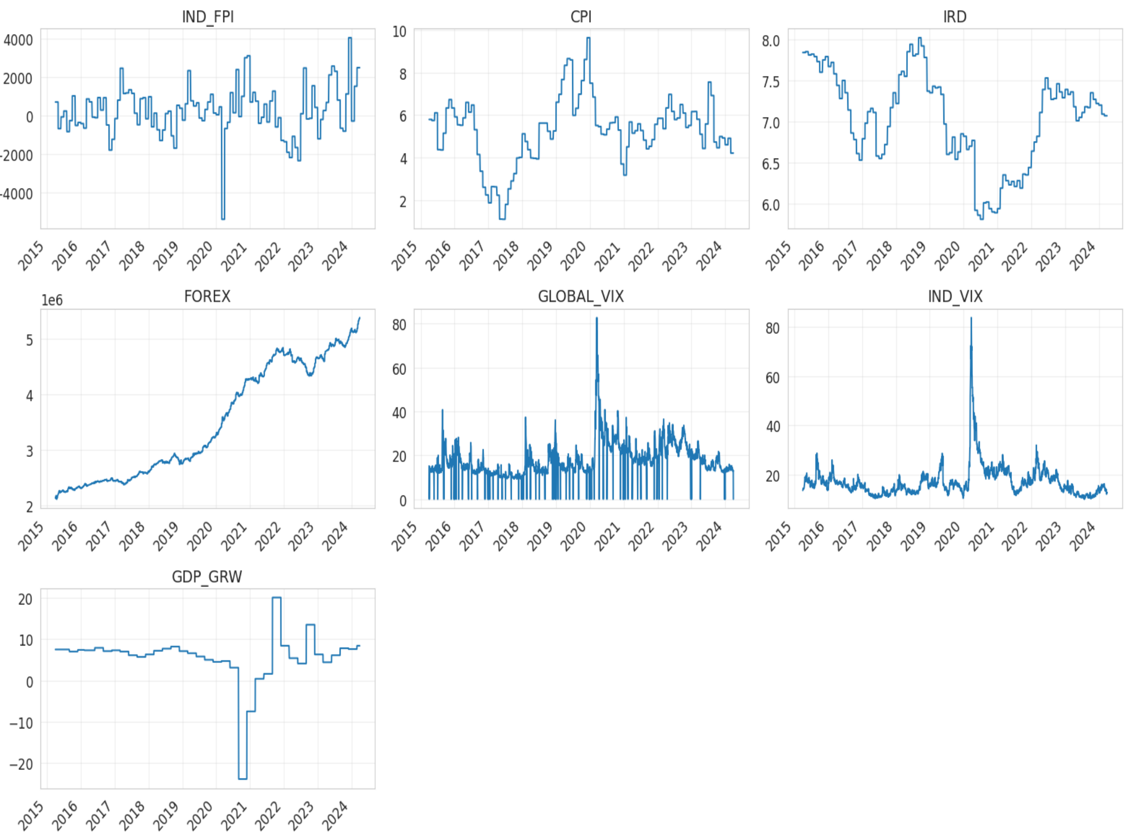

Figure 5. Panel of all the important variables

Source: Author's own representation in Python by using the data available in RBI, NSE, MOSPI, CBOE VIX, NSDL from 2015-2024